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1 WITH 15-MINUTE RAPID CHARGING TECH, EXPONENT EYES COMMERCIAL EVs

Exponent Energy (Exponent), with its 15-minute rapid charging tech, aims to help the electric vehicle OEMs launch and sell economically priced electric vehicles. The startup says that its patented 'water-based' off-board thermal management system can charge an electric three-wheeler equipped with batteries provided by Exponent from 0 to 100% in 15 minutes



e/pump and e/pack

Also undergoing a trial with a major commercial bus manufacturer, the startup says it will launch the tech for electric bus OEMs soon. However, it has no immediate plans to venture into the two-wheeler segment, Arun Vinayak, Co-founder & CEO, told Electronics For You during an exclusive conversation. Once launched, the tech will charge electric buses from 0 to 100% in 15 minutes.

He says, "We don't differentiate in vehicle forms from a technology

point of view. The energy consumption by two-wheelers deployed in the logistics segment is much lesser than that consumed by three- and four-wheelers. The question we want to answer is not how many electric vehicles we are helping in charging rapidly, but how much energy we can save while doing the same."

"Two years from now, we might enter the two-wheelers segment. But for now, we know that most Indian startups don't die of starvation, but they die of indigestion," he adds.

What this means for commercial EVs. According to Vinayak, the biggest hurdle in India's race towards EV adoption is the efficiency of these vehicles. He points out towards degradation of battery life and slow charging capabilities as two factors hindering the growth. In numbers, fast charging generated 246 times more heat than the industry average of 4-hour charging.

"If you rapidly charge, your battery life decreases, and chances of battery mishaps increase. We realised that charging a battery as fast as we can was not the only problem statement, but making sure that the health of batteries remains positive even after rapid charging was," he explains.

TUV India (TUV Nord Group, Germany) tests found only 13% degradation after 3,000 charging cycles of 15-minute rapid charging. Vinayak noted that even in real-life conditions, this deterioration would not be any bigger than 16% to 18%.

In the real world, this would mean that the 3,000-cycle warranty can make EVs apply for five-year financing (most commercial EVs in India cannot be financed for over three years at the moment). Moreover, it will allow commercial EV drivers to spend maximum time on the road, a prerequisite for ensuring profit in the logistics industry.